

# SUSHANT GARGI

Chennai, Tamil Nadu

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## Professional Summary

Full-Stack Engineer building agentic AI systems and microservices with multi-agent orchestration, Model Context Protocol (MCP), and LLM coordination. Engineered AI-powered automation and CI/CD pipelines reducing manual testing by 40% and test authoring by 32%. Expert in Spring Boot, React, PostgreSQL/Redis, Kubernetes, and autonomous agent workflows with human-in-the-loop governance.

## Education

### Vellore Institute of Technology

B.Tech in Computer Science and Engineering (specialization in AI and Robotics); CGPA: 8.78/10.0

Aug. 2021 – Jul. 2025

Chennai, Tamil Nadu

### Taxila Public School

Higher Secondary Education (12th Grade); Percentage: 96.4%

2021

Meerut, Uttar Pradesh

## Professional Experience

### COGNIZANT

Apr. 2025 – Present

Junior Software Development Engineer in Test (SDET)

Chennai, India

- Developed scalable UI automation frameworks for enterprise and desktop applications using Page Object Model-based Selenium, TestNG, Apache POI, and TestComplete. Reduced manual testing effort by 40% and improved regression accuracy by 35%.
- Implemented containerized CI/CD pipelines via Docker for reproducible QA workflows, integrating Git, Maven, Jenkins, and Selenium Grid for parallel execution across 8 test suites to reduce regression cycle time by 25%, while delivering end-to-end REST API validation using Postman and REST Assured.
- Partnered with the GenAI team to automate test case generation and data setup, cutting authoring time by 32% per testing cycle, and authored behavior-driven test scenarios with Cucumber and Gherkin to align QA processes with Agile user stories across product releases.

### ETHNUS

Aug. 2023 – Nov. 2023

MERN Full Stack Internship Program

Remote

- Built comprehensive REST APIs for hotel search, booking workflows, rate calculation, and customer account operations using Node.js and Express serving production-level request volumes, while establishing JWT authentication with RBAC to secure critical endpoints and reduce unauthorized access by 30%.
- Designed 12+ responsive React UI components for search, booking flows, dashboards, and admin operations. Implemented Stripe payment integration with real-time webhooks for booking confirmations, inventory synchronization, and 500+ concurrent transactions.
- Optimized MongoDB database performance through advanced schema design and indexes, successfully improving average API response latency by 35%.

## Projects

**Lovable Clone (Vibe Coding Tool)** | Java, Spring Boot, React, Redis, PostgreSQL, Docker, Kubernetes, Spring AI | [Link](#)

- Established a scalable Spring Boot backend and React frontend utilizing MVC architecture to support 100+ concurrent users. Delivered secure REST APIs across 7 controllers (Auth, Projects, Chat, Files, Billing, Members, Usage) with JWT/RBAC for multi-tenant isolation.
- Integrated Spring AI LLM with streaming responses and custom advisors, accelerating performance by 35% via Redis caching and rate limiting, validated through JUnit with 13+ service implementations.
- Deployed Kubernetes microservice architecture with Docker, integrating MinIO, PostgreSQL, and Redis across 14 entities and 10 repositories for multi-environment, highly available configurations.

**AirBnB Clone (Hotel Booking & Pricing API)** | Java, Spring Boot, Spring Security, JPA, JWT, Stripe | [Link](#)

- Constructed 5 scalable Spring Boot microservices for property discovery, transactional booking flows, inventory, pricing, and user management handling 1,000+ daily bookings. Incorporated a surge-based pricing algorithm via the Strategy design pattern for occupancy, holiday, and urgency-based optimization.
- Automated payment processing utilizing the Stripe payment gateway and real-time inventory updates via webhooks for 15+ API endpoints. Secured all endpoints with JWT authentication, role-based access control (RBAC), global exception handling, and standardized API responses.

**Human Interpretable Credit Card Fraud Detection System** | Python, PyTorch, KAN (pykan) | [Link](#)

- Developed LTACNN-KAN framework achieving 99.7% accuracy on 284,807 transactions for interpretable credit card fraud detection.
- Achieved 0.995+ Precision/Recall/ROC-AUC via KAN, surpassing LTACNN baseline in interpretability.
- Validated model transparency using KAN across 30 features via SHAP/LIME analysis (Wilcoxon  $p > 0.05$ ).

## Technical Skills & Certifications

**Full Stack Development:** Java, Python, JavaScript, TypeScript, SQL, HTML5, CSS3, Spring Boot, Spring Security, Spring AI, JPA/Hibernate, Node.js, Express.js, Microservices, REST APIs, JWT, OAuth2, React.js, Next.js, Tailwind CSS, Responsive Design

**AI/ML & Data Engineering** RAG, LangGraph, LangChain, Prompt Engineering, LLM Orchestration, Agentic AI, Vector Search, PyTorch, TensorFlow, Hugging Face, pandas, NumPy, PostgreSQL, MongoDB, MySQL, Redis, Kafka, AWS S3

**Testing, DevOps & Cloud Infrastructure:** Selenium, TestNG, Cucumber, REST Assured, Postman, JUnit, API Testing, UI Automation, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD Pipelines, Maven, Git, AWS, GCP

**Certifications:** [Automation Pro I \(Workato\)](#), [Google Cloud Computing Foundations](#)